

LSM 6DSO IMU 1 | SPI 2

- User Bank 0 Registers

Addr (Hex)	Register Name	Data
01	FUNC_CFG_ACCESS	
0D	INT1_CTRL	0b11
0F	WHO_AM_I	
10	CTRL1_XL	0b10000100
11	CTRL2_G	0b10001100
12	CTRL3_C	0b01000100
14	CTRL5_C	
17	CTRL8_XL	0b0
19	CTRL10_C	0b100000
1E	STATUS_REG	
20	OUT_TEMP_L	TEMP_DATA[7:0]
21	OUT_TEMP_H	TEMP_DATA[15:8]
22	OUTX_L_G	GYRO_DATA_X[7:0]
23	OUTX_H_G	GYRO_DATA_X[15:8]
24	OUTY_L_G	GYRO_DATA_Y[7:0]
25	OUTY_H_G	GYRO_DATA_Y[15:8]
26	OUTZ_L_G	GYRO_DATA_Z[7:0]
27	OUTZ_H_G	GYRO_DATA_Z[15:8]
26	OUTX_L_A	ACCEL_DATA_X[7:0]
29	OUTX_H_A	ACCEL_DATA_X[15:8]
2A	OUTY_L_A	ACCEL_DATA_Y[7:0]
2B	OUTY_H_A	ACCEL_DATA_Y[15:8]
2C	OUTZ_L_A	ACCEL_DATA_Z[7:0]
2D	OUTZ_H_A	ACCEL_DATA_Z[15:8]
40	TIMESTAMP0	TIMESTAMP_DATA[31:24]
41	TIMESTAMP1	TIMESTAMP_DATA[23:16]

Addr (Hex)	Register Name	Data
42	TIMESTAMP2	TIMESTAMP_DATA[15:8]
43	TIMESTAMP3	TIMESTAMP_DATA[7:0]
56	TAP_CFG0	0b01000001

• Characteristics

- Gyroscope
 - Gyroscope range: 2000 dps
 - Sensitivity for 2k dps: 70 mdps/LSB
 - Initial zero rate output tolerance: +- 1 dps
 - ODR: 1666 Hz
- Accelerometer
 - Accelerometer range: 16 g
 - Sensitivity for 16 g: 0.488 mg/LSB
 - Initial zero rate output tolerance: +- 20 mg
 - ODR: 1666 Hz
- Temperature
 - Sensitivity: 256 LSB/C

• SPI

- Pull CS to activate: LOW
- Data delivered MSB first LSB last
- Data transitioned on falling edge of SCLK

• Self test at boot for later version

• Interrupts

- INT fire pull: HIGH
- Use INT 1
- INT fired when DATA READY, cleared when INT STATUS bit READ

• Time stamp of reading stored in register, resolution of 25 us

• Power mode: 6-axis with independant ODR

• Block data update